Kohn-Sham model of dirac16complex in the author's primordial field with the deflating extra times

The block reduction, the instantaneous ground and first excited states along the prescribed history $a_4 = AHx_4$, Mermin thermodynamics, adiabaticity, the energy-momentum tensor, the rescaling partners and the source conditions, with the verification records and an exact statement of what is computed and what is not

dirac16complex project (with Claude Opus 5.5)

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Abstract

This document records the Kohn-Sham (density-functional) model of the fermion field dirac16complex in the author's primordial gravitational field, deliverable 4 of `Revision/SPEC.md` section 10, as computed by the code in `Revision/kohn_sham/`. Exact (two independent verifiers, WolframScript and sympy): in the good sector (no extra-time momentum) the ansatz $\Psi = e^{ik\cdot x}W^{-3}\chi(y, x_4)$ removes the spin-connection term $3H\gamma^0$ and reduces the 16-component equation, for every history $a_4(x_4)$, to eight 2×2 blocks $h_j = j[-i\sigma_1\, d/dy + M_{\text{eff}}\sigma_2 + \kappa k\sigma_3] + v_v$ in the hidden coordinate y, with $\kappa = e^{-Hy-a_4}$; the deflation of the extra times enters the instantaneous problem only through this redshift of the 3-momenta and through the extra-time volume; the exact local exchange of the uniform good-sector gas gives $M_{\text{eff}} = m + \frac{15}{16}\lambda S$ and $v_v = -\lambda n/16$. Computed (a Rust solver, an independent Python reference with a different discretisation, and a cross-check of the full canonical matrix, 31 of 31 checks PASS over 162691 comparisons): the instantaneous ground and first excited states of 75 parameter sets ($N = 8, 136, 688$; five couplings; five slices $a_{4,0}$ of the prescribed history $a_4 = AHx_4$, $A = 1$), 135 Mermin states at three temperatures, the adiabaticity measure ($Q_{\max} \leq 0.09345324$ over the matrix), the Fermi-level crossings (none in the canonical matrix; a demonstration at $N = 696$), the energy-momentum tensor profiles, the exact rescaling partners and the exact-exchange variant. ASSUMED or chosen: the Z2 brane at the patch end, the regular tip at $y = -3$, the filling convention. Measured afterwards (`Revision/kohn_sham/tip_convergence/`, $L = 3$ to 6, 7 of 7 checks PASS): the free results converge as the tip cutoff L grows and the recorded $L = 3$ values of E_{KS} lie within 1.4% of the large-L values, except $N = 688$ at $a_{4,0} = 0$, which changes its occupied set from $L = 3.5$ on and does not converge up to $L = 6$; the interaction energy of the brane zero modes depends strongly on L (the recorded $N = 8$ value is about one third of the large-L value), and for 7 of the 19 interacting states studied (self-consistent iteration failing at larger L before the differences settle) the limit $L \to \infty$ of E_{KS} is not established. Not admissible as sources: no recorded Kohn-Sham state satisfies the source conditions of the a_4 equations, so the history is a prescribed background without back-reaction. Not done: the time-dependent (non-adiabatic) Kohn-Sham problem, correlation beyond exchange, back-reaction.

1 The task and the answer

The author asked (2026-10-01, quoted from `Revision/README.md`): "plan and employ a DFT-motivated approximation similar to the one that you already created [...], with a Kohn-Sham fermion-gas thermodynamic effective potential, that employs the DFT ground and first-excited-state computation". `Revision/SPEC.md` section 7 makes this precise: a Kohn-Sham fermion-gas model of dirac16complex in the good sector with the contact interaction $U = \frac{\lambda}{2}S^2$, the Hartree mean field plus the exact local exchange of the uniform gas, Mermin's finite-temperature functional, the reduction to 2×2 blocks in the hidden coordinate, the boundary conditions, the ground and first excited states, the thermodynamics, and, because a_4 depends on x_4, the INSTANTANEOUS (adiabatic) Kohn-Sham states along the history with the extra times deflating, with an adiabaticity measure.

The answer of this record:

1. Exact (section 3, section 4): the block reduction, the exchange functional, the boundary conditions, the rescaling identity, the energy-momentum tensor and its conservation laws, the adiabaticity measure. Every formula is in `Revision/kohn_sham/ks-theory.json` and verified twice, in `Revision/kohn_sham/reports/ks-theory-wolfram.json` (46 of 46 PASS) and independently in `Revision/kohn_sham/reports/ks-theory-python.json` (58 of 58 PASS).

2. Computed (sections 5 to 10): the instantaneous Kohn-Sham states of the canonical matrix by the Rust solver, independently by the Python reference, and compared state by state on the full matrix.

3. Not admissible (section 11): the recorded states are not sources of the field equations for a_4; the history $a_4 = AHx_4$ is PRESCRIBED.

4. Open (section 15): the time-dependent Kohn-Sham problem, correlation, back-reaction.

Every number below is read from the named Revision report or results file (the publication test re-reads them); units: $H = m = 1$, energies, momenta and temperatures in units of m, as in `Revision/kohn_sham/results/parameters.json`. Interpretations are labelled.

2 Setting: the author's field, the good sector and the hidden coordinate

The coordinates are the author's, $x_0, x_1, \ldots, x_7$, in the author's order: x_0 is the hidden space direction, $z = 6Hx_0 \in (0, \pi/2)$; x_1, x_2, x_3 are ordinary 3-space, inflating with the scale factor $e^{a_4} \sin^{1/6} z$; x_4 is the time; x_5, x_6, x_7 are the three extra times, which DEFLATE exponentially, with the scale factor $e^{-a_4} \sin^{1/6} z$ and a_4 increasing (they are never treated as static). (Until 2026-10-14 this record named them $x_1, \ldots, x_8$, with the hidden direction x_8 listed last; the user ordered the author's coordinates on 2026-10-14.) The metric of `Revision/SPEC.md` section 1, as changed by the author on 2026-10-14 ($g_{55} = g_{66} = g_{77} = -e^{-2a_4(x_4)}\, b_4(x_4) \sin^{1/3}(6Hx_0)$; `Revision/metric_b4/METRIC_CHANGE_PROVENANCE.md`), is

$$\begin{aligned} ds^2 = {} & \cot^2 z\, dx_0^2 + e^{2a_4(x_4)} \sin^{1/3} z\, (dx_1^2 + dx_2^2 + dx_3^2) - dx_4^2 \\ & - e^{-2a_4(x_4)}\, b_4(x_4) \sin^{1/3} z\, (dx_5^2 + dx_6^2 + dx_7^2); \end{aligned}$$

$b_4 = 1$ is the metric used before 2026-10-14, and every number of this record is computed at $b_4 = 1$, which is the vacuum background (with the author's weights every vacuum solution has b_4 constant and $a_4 = \pm Hx_4 + \text{const}$, `Revision/metric_b4/results/vacuum-b4.json`). Section 2.1 shows that the good-sector Kohn-Sham problem does not depend on b_4 at all.

The Kohn-Sham record uses the hidden coordinate $y = \ln(\sin z)/(6H) \in (-\infty, 0]$, with $dy = \cot z\, dx_0$; the patch end $z = \pi/2$ is $y = 0$ (the brane) and the tip $z \to 0$ is $y \to -\infty$. In it the metric is exactly

$$ds^2 = e^{2Hy}\big[e^{2a_4}(dx_1^2 + dx_2^2 + dx_3^2) - e^{-2a_4} b_4 (dx_5^2 + dx_6^2 + dx_7^2)\big] - dx_4^2 + dy^2,$$

with the warp $W = e^{Hy}$ and $\sqrt{|g|} = b_4^{3/2} e^{6Hy}$ ($= b_4^{3/2} \cos z$ in the author's chart; $= e^{6Hy}$ at $b_4 = 1$). The 7-volume is independent of a_4: the inflation e^{3a_4} of 3-space is compensated exactly by the deflation e^{-3a_4} of the extra times; it carries the factor $b_4^{3/2}$ of the extra-time components (equal to 1 on the background $b_4 = 1$ of this record). The canonical spin connection gives $\gamma^\mu \Omega_\mu = 3H\gamma^0$ for every $a_4(x_4)$ at $b_4 = 1$ (for general b_4 it is $3H\gamma^0 + \frac{3}{4}(b_4'/b_4)\gamma^4$, section 2.1): the a_4' pieces of the three inflating directions, $+\frac{3}{2}a_4'\gamma^4$, and of the three deflating directions, $-\frac{3}{2}a_4'\gamma^4$, cancel.

The good sector: no dependence on x_5, x_6, x_7 (zero extra-time momenta), a coordinate 3-torus of side ℓ with the momenta $k \in \Delta k\,\mathbb{Z}^3$, $\Delta k = 2\pi/\ell$, and the extra-time coordinate volume v_t; $\mathrm{Vol}_7 = \ell^3 v_t$. With the ASSUMED brane condition and the chosen tip condition (section 5; together they kill the y-current and make the blocks h_j self-adjoint) the good sector is quantised in the positive realisation of the canonical quantisation of `Revision/SPEC.md` section 6; the field-theory record `Revision/docs/DIRAC16COMPLEX_FIELD_THEORY` (its section 11) constructs that realisation explicitly only for single good-sector momenta with frozen coefficients, and without a boundary condition at $z = \pi/2$ the curved good sector has complex frequencies (for $U = 0$ whenever $m^2 < 9H^2$, which includes the parameters $m = H = 1$ used here). The expectation-value rule $\langle \Psi^\dagger M \Psi \rangle = \mathrm{Tr}(M\rho)$, $\rho = \sum_{\mathrm{occ}} f\, uu^\dagger B$, defines the quasi-free (Slater or Mermin) states used here.

item	Wolfram records	sympy records
gammas, C, B, Γ	`fixture_input`, `clifford_relation`, `C_B_Gamma`	`fixture_input`, `fixture_python_equals_wolfram`, `clifford_relation`, `C_B_Gamma`, `BC_and_Cg4`
hidden coordinate, $\sqrt{\lvert g\rvert}$	`geometry_hidden_coordinate`, `geometry_sqrt_det`	`geometry_hidden_coordinate`, `geometry_sqrt_det`, `geometry_warped_form`
spin connection	`spin_connection_compatibility`, `spin_connection_slash_3H`, `spin_connection_time_terms_cancel`, `spin_connection_slash_3H_x0_chart`	`spin_connection_compatibility`, `spin_connection_slash_3H`, `spin_connection_time_terms_cancel`, `spin_connection_time_term_value`, `spin_connection_slash_3H_x0_chart`

2.1 The metric of 2026-10-14 (b4)

On 2026-10-14 the author changed the extra-time components to $g_{55} = g_{66} = g_{77} = -e^{-2a_4(x_4)}\, b_4(x_4) \sin^{1/3}(6Hx_0)$ ($b_4 = 1$ is the earlier metric; the change, its source cells and every verification are recorded in `Revision/metric_b4/METRIC_CHANGE_PROVENANCE.md`). The report `Revision/kohn_sham/reports/ks-b4-wolfram.json` (4 of 4 PASS, producer `Revision/kohn_sham/theory/verify_ks_b4.wls`) proves exactly, for every 16-component $X(x_0, \ldots, x_7)$:

1. Factorisation of the Dirac operator (record `K1_dirac_operator_b4_factorisation`):

$$D_{b_4}\big[b_4^{-3/4} X\big] = b_4^{-3/4}\Big(D_1[X] + \big(b_4^{-1/2} - 1\big)\, e^{a_4} \sin^{-1/6} z \sum_{t=5}^{7} \gamma^t \partial_t X\Big),$$

where $D_{b_4} = \gamma^\mu D_\mu$ in the metric with b_4 and D_1 the same operator at $b_4 = 1$.

1. The good sector is independent of b_4 (record `K2_good_sector_independent_of_b4`): for X independent of x_5, x_6, x_7 (no extra-time momenta) the extra term vanishes, $D_{b_4}[b_4^{-3/4} X] =$

$b_4^{-3/4} D_1[X]$ exactly; the χ-equation, the slice Hamiltonian $h(a_{4,0})$, every level, orbital and slice of this record are unchanged.

2. Norm and volume (record `K3_norm_and_volume`): $\sqrt{|g|} = b_4^{3/2} \cos z$ and $\sqrt{|g|}\,(b_4^{-3/4})^2 = \cos z$, so the factor $b_4^{-3/4}$ keeps every normalisation integral; the proper extra-time volume carries $b_4^{3/2}$, and the rescaling identity of section 10 reads

$$\mathrm{KS}(a_{4,0}, b_{4,0};\, \Delta k, v_t) = \mathrm{KS}(0, 1;\, \Delta k\, e^{-a_{4,0}},\ v_t\, e^{-3a_{4,0}}\, b_{4,0}^{3/2}).$$

1. The background (record `K4_background_b4_1`): $\gamma^\mu \Omega_\mu = 3H\gamma^0 + \frac{3}{4}(b_4'/b_4)\gamma^4$ in the metric with b_4, and $= 3H\gamma^0$ at $b_4 = 1$, the value `ks-theory.json` uses. The vacuum history of `Revision/metric_b4/results/vacuum-b4.json` has $b_4 = 1$ (every vacuum solution with the author's weights has b_4 constant, and a constant b_4 is absorbed into v_t by item 3), so on the background of this record no recorded number changes.

The source conditions of section 11 hold for every b_4 (record `ks_verdict_holds_for_every_b4` of `Revision/field_equations_a4/reports/ks-source-conditions.json`): the left-hand sides of the diagonal a_4 equations stay free of x_0, so C1 is unchanged and the x_0-dependent profiles are not an admissible source for any b_4; C2 as written is its $b_4 = 1$ form; the off-diagonal x_4-x_0 condition $\kappa q_{40} = \cot z\,(b_4'/b_4)\,Q$ is not evaluated, because the solver does not record $T^{x_4}{}_{x_0}$ (OPEN). Not established here: Kohn-Sham states with nonzero extra-time momenta, for which the term $(b_4^{-1/2} - 1)$ of item 1 does not vanish, and any history with non-constant b_4.

report	**records**
`Revision/kohn_sham/reports/ks-b4-wolfram.json`	`K1_dirac_operator_b4_factorisation`, `K2_good_sector_independent_of_b4`, `K3_norm_and_volume`, `K4_background_b4_1`

3 The exact block reduction

Ansatz. $\Psi = e^{ik\cdot x} W^{-3} \chi(y, x_4)$ gives EXACTLY, for a time-dependent a_4,

$$\gamma^0 \partial_y \chi + \gamma^4 \partial_{x_4} \chi + i\kappa\, k \cdot \gamma\, \chi = (M_{\mathrm{eff}} - i v_v \gamma^4)\chi, \qquad \kappa(y, x_4) = e^{-Hy - a_4(x_4)},$$

i.e. $i\partial_{x_4}\chi = h(x_4)\chi$ with h Hermitian in $\int dy\, \chi^\dagger \chi$. The factor W^{-3} removes the term $3H\gamma^0$; without it the term survives. At a fixed instant the instantaneous (adiabatic) problem is $h(a_{4,0})\chi = \varepsilon\chi$ with $a_{4,0} = a_4(x_4)$.

Blocks. $J = \gamma^0\gamma^1\gamma^4$ ($J^2 = 1$), $K_1 = \gamma^2\gamma^3$ and $K_2 = \gamma^5\gamma^6$ ($K^2 = -1$) commute with each other and with γ^0, $\gamma^0\gamma^1$, $\gamma^0\gamma^4$, B and C; their joint eigenspaces, labelled (j, s_2, s_3) with $j, s_2, s_3 = \pm 1$, are eight rank-2 blocks with an explicit unitary basis V (entries of $2\sqrt{2}\,V$ in $\{0, \pm 1, \pm i\}$). For k along x_1 (the spinor rotation commutes with γ^0, γ^4, B, C, so the spectra depend on $|k|$ only), every block of type j carries

$$h_j = j\big[-i\sigma_1 \tfrac{d}{dy} + M_{\rm eff}(y)\sigma_2 + \kappa(y)k\sigma_3\big] + v_v(y),$$

equivalently $\chi' = N\chi$ with $N = M_{\rm eff}\sigma_3 - \kappa k\sigma_2 + ij(\varepsilon - v_v)\sigma_1$; in the real form $\chi = (a, ib)$, $a' = M_{\rm eff}a - (\kappa k + j(\varepsilon - v_v))b$ and $b' = (j(\varepsilon - v_v) - \kappa k)a - M_{\rm eff}b$. In the block basis $\gamma^0 = \sigma_3$, $B = js_2$ and $C = s_2\sigma_2$.

Symmetries and degeneracies. $h_{-1} - v = -(h_{+1} - v)$ and $\sigma_3 h_j(k)\sigma_3 = h_{-j}(-k)$: the (j,k) and $(-j,-k)$ orbitals are degenerate. On a lattice shell $|k|^2 = \Delta k^2 n_2$ with $r_3(n_2)$ vectors every level of $h_j(|k|)$ is $4r_3(n_2)$-fold for each j; at $k = 0$ every level is 4-fold per j. The chirality Γ maps block (j, s_2, s_3) onto $(-j, s_2, s_3)$ with the 2×2 entry $s_2\sigma_2$, and $\sigma_2 h_j(M,k)\sigma_2 = h_{-j}(-M,k)$ with the brane parities exchanged (the ingredient of T3, section 12).

Where the deflation enters (exact). $a_{4,0}$ enters h_j only through $\kappa k = e^{-Hy}(ke^{-a_{4,0}})$: the 3-momenta redshift as ke^{-a_4}. Because the good sector has no extra-time momentum, the extra-time scale factor e^{-a_4} enters the instantaneous problem only through the extra-time volume (section 10).

item	Wolfram records	sympy records
ansatz and Hamiltonian	`ansatz_removes_spin_connection`, `ansatz_without_W3_term_survives`, `hamiltonian_16_hermitian`	`ansatz_removes_spin_connection`, `ansatz_without_W3_term_survives`, `hamiltonian_16_hermitian`
blocks	`blocks_commuting_set`, `blocks_basis_unitary`, `blocks_forms`, `blocks_relation_to_Gamma`, `block_hamiltonian`, `block_ode_equivalent`	`blocks_commuting_set`, `blocks_projectors`, `blocks_basis_unitary`, `blocks_forms`, `blocks_not_everything_block_diagonal`, `blocks_relation_to_Gamma`, `block_hamiltonian`, `block_ode_equivalent`
symmetries	`block_type_relation`, `block_Gamma_map`, `rotation_invariance`	`block_type_relation`, `block_Gamma_map`, `rotation_invariance`
solver (numerical basis)		`block_reduction_numeric`, `gamma_fixture_numeric` (Rust solver report)

4 The functional: Hartree, exact uniform-gas exchange, Mermin

Exchange (exact for the uniform gas). The interaction is $U = \frac{\lambda}{2}S^2$, $S = \bar\Psi\Psi$, normal ordered ($\lambda > 0$ repulsive). For a quasi-free state, Wick's theorem gives $\langle : S^2 : \rangle = (\mathrm{Tr}\, C\rho)^2 - \mathrm{Tr}(C\rho C\rho)$. For every $p \to -p$ symmetric occupation of the 8-fold good-sector gas, at every temperature, $\rho = (nB + SC)/16$ and

$$e_H = \tfrac{\lambda}{2}S^2, \qquad e_x = -\tfrac{\lambda}{32}(n^2 + S^2), \qquad e_{\rm int} = \tfrac{15}{32}\lambda S^2 - \tfrac{1}{32}\lambda n^2,$$

$$M_{\text{eff}}(y) = m + \tfrac{15}{16}\lambda S(y), \qquad v_v(y) = -\tfrac{\lambda}{16} n(y),$$

with proper densities; for one filled level at rest $E_x/E_H = -1/8$, and on shell $\langle\mathcal{L}\rangle/\sqrt{|g|} = e_{\text{int}}$. No correlation is included (Hartree plus exchange only). The Rust solver reads the coefficients from `ks-theory.json` and checks them (`theory_input_coefficients`: $M_{\text{eff}} = m + 0.9375\lambda S$, $v_v = -0.0625\lambda n$).

Exact Fock exchange of the slab (diagnostic and variant). For closed shells the exact Fock exchange of the Kohn-Sham determinant is also local, $e_x^{\text{exact}} = -\frac{\lambda}{32}(n^2 + S^2 - Q^2 - Y^2)$ with $Q = \langle\Psi^\dagger B\gamma^0\Psi\rangle$ and $Y = 0$; the canonical functional is the uniform-gas one, the difference $\Delta E_x = \frac{\lambda}{32}\,2\text{Vol}_7 \int e^{6Hy} Q^2\, dy$ is reported for every state, and the exact-Fock functional is solved as a variant (60 states, `Revision/kohn_sham/results/exx/exact-fock-variant.csv`).

Densities, energy, Mermin. Per orbital, with $P(y) = e^{-6Hy}/(\ell^3 v_t)$: $n_o = P\chi^\dagger\chi$, $s_o = P\chi^\dagger j\sigma_2\chi$, $t_o = P\chi^\dagger j\sigma_3\chi$, $q_o = P\chi^\dagger\sigma_3\chi$. Totals are $\sum w\, g\, f$ times these, $g = 4r_3(n_2)$, $w = 1/2$ (section 5). The Kohn-Sham energy of the doubled system is $E_{KS} = \sum gf\varepsilon - 2\text{Vol}_7 \int_{-L}^{0} e^{6Hy} e_{\text{int}}\, dy$; Mermin's occupations are $f = 1/(1 + e^{(\varepsilon-\mu)/T})$ with μ fixed by $\sum gf = N$, the entropy is $S_{\text{ent}} = -\sum g[f\ln f + (1-f)\ln(1-f)]$, the grand potential (the thermodynamic effective potential) is $\Omega = -T\sum g\ln(1 + e^{-(\varepsilon-\mu)/T}) - 2\text{Vol}_7 \int_{-L}^{0} e^{6Hy} e_{\text{int}}\, dy$, and $F = \Omega + \mu N$.

Discrepancy recorded, not hidden: the line "mu fixed by sum w_Z2 g f = N" of `ks-theory.json` is inconsistent with its own densities, energy and grand potential; the solver and the reference use $\sum gf = N$, and `Revision/kohn_sham/results/parameters.json` records the discrepancy.

item	Wolfram records	sympy records
uniform gas and exchange	`gas_mode_projector`, `gas_angular_average`, `gas_densities`, `exchange_uniform_gas`	`gas_mode_projector`, `gas_angular_average`, `gas_negative_energy_modes`, `gas_densities`, `hf_wick_contraction`, `exchange_uniform_gas`, `filled_shell_ratio`
Kohn-Sham potentials	`ks_potentials`, `ks_onshell_lagrangian`	`ks_potentials`, `ks_onshell_lagrangian`, `ks_theory_json_exchange`
exact Fock exchange of the slab	`exchange_slab_exact_fock`	`exchange_slab_exact_fock`

5 Boundary conditions, filling convention, numerics and parameters

Brane (ASSUMED). The Z2 mirror (orbifold) at $y = 0$: $\Psi(-y) = \pm\gamma^0\Psi(y)$ with the mirror warp $e^{-H|y|}$; continuity gives $(1 \mp \gamma^0)\chi(0) = 0$, i.e. the even parity $\chi_2(0) = 0$ ($b(0) = 0$) and the odd parity $\chi_1(0) = 0$ ($a(0) = 0$). Both conditions kill the y-current and make h_j self-adjoint. The mirror map P_A takes $(M(y), v(y))$ to $(-M(-y), v(-y))$: the doubled problem is P_A-symmetric exactly when the mirror copy carries $(-m, +\lambda)$. Both parities are solved and filled together; N

is the particle number of the doubled system (universe and Z2 image), the patch holds $N/2$, and $w = 1/2$ per patch.

Tip (chosen). At $y = -L$: $(1 - Q(\theta))\chi(-L) = 0$, $Q(\theta) = \cos\theta\,\sigma_3 + \sin\theta\,\sigma_2$, canonical $\theta = 0$, cutoff $L = 3$. For $k \neq 0$ an orbital is suppressed toward the tip like $\exp(-|k|(\kappa(-L) - \kappa(y))/H)$. The dependence on L is measured in `Revision/kohn_sham/tip_convergence/` ($L = 3$ to 6 in steps of 0.5 with the record's step $h = 1/300$, every run repeated at $h/2$; a copy of the solver that reads L from the environment reproduces the committed matrix byte for byte at $L = 3$; the reference solver agrees at $L = 3$ and 4; report `Revision/kohn_sham/tip_convergence/tip-convergence.json`, 7 of 7 checks PASS). The free results with $k \neq 0$ converge faster than exponentially (this suppression), and the recorded $L = 3$ values are low by up to 1.4% in E_{KS} and 12% in $2\mathrm{Vol}_7 \int e^{6Hy} p_0\, dy$ ($N = 136$, $\lambda = 0$, $a_{4,0} = 2$: 12.44507 against the extrapolated 12.62207, and 8.47270 against 9.63839). The $k = 0$ bulk levels approach $\pm m$ only like $(n\pi)^2/(2mL^2)$, so $N = 688$ at $a_{4,0} = 0$ does not converge (Parameters below). The brane zero modes have a proper density that grows toward the tip like $e^{(6H-2m)|y|}$ (exact, verified), so the interacting $N = 8$ energy depends strongly on L: $E_{KS} = \mp 9.868 \times 10^{-4}$ at $L = 3$ ($\lambda = \pm\lambda_1$, every slice) against $\mp 3.0046 \times 10^{-3}$ at large L (extrapolated from $\lambda = -\lambda_1$ at $a_{4,0} = 2$, the only interacting $N = 8$ state solved up to $L = 6$; the $\pm\lambda_1$ partners are negatives of each other to 1.1×10^{-12} relative at every L where both converge; the other $N = 8$ states extrapolate to $\mp 3.007 \times 10^{-3}$ from $L \leq 4.5$, because their self-consistent iteration fails from $L = 5$): the recorded value is about one third of the large-L value. The self-consistent iteration fails at some larger L for 13 of the 19 interacting states studied (a failure of the iteration, not a proof that no self-consistent state exists); for 7 of them the successive differences of E_{KS} do not meet the geometric-tail rule of the study before it fails, so their limit is not established.

Exact $k = 0$ spectra. For constant $M > 0$, $v = 0$, $\theta = 0$: even parity $\varepsilon = 0$ with the brane zero mode $\chi = (e^{My}, 0)$, and $\varepsilon = \pm\sqrt{M^2 + (n\pi/L)^2}$; odd parity $\varepsilon = \pm\sqrt{M^2 + p^2}$ with $\tan(pL) = -p/M$. The brane band has the slope $d\varepsilon/dk = jc$ at $k = 0$, $c = e^{-a_{4,0}} \frac{2M}{2M-H} \frac{1-e^{-(2M-H)L}}{1-e^{-2ML}}$; for $M = H = 1$, $L = 3$: $c = 2/(1 + e^{-3}) = 1.905148253644866$ (record `brane_band_slope`).

Filling (CONVENTION, justification OPEN). Particles occupy the positive branch (the levels with $\varepsilon > 0$ of the $\lambda = 0$ problem) and the $k = 0$ brane zero modes, each followed continuously in λ; the negative branch is the normal-ordered sea and contributes nothing to N, n, S or the energy-momentum tensor. The excluded thermal sea holes are reported as a diagnostic (section 8).

Numerics. The Rust solver (`Revision/kohn_sham/solver`, own crate, no external crates) solves each block by shooting with RK4 on 900 steps, Pruefer counting and a root tolerance of 1e-13, Anderson-mixed SCF to 1e-11. The independent Python reference (`Revision/kohn_sham/reference`) uses a staggered finite-difference discretisation on three grids (300, 600, 1200) with Richardson extrapolation and a stated uncertainty, validated on a fourth grid; its convergence order is two (record `free_convergence_order_two`).

Parameters (`Revision/kohn_sham/results/parameters.json`): $H = m = 1$, $L = 3$, $\Delta k = 0.25$, $v_t = 1$, $\mathrm{Vol}_7 = 15875.21366031351$, tip $\theta = 0$, history $A = 1$, slices $a_{4,0} \in \{0, 0.5, 1, 1.5, 2\}$, temperatures $T \in \{0.01, 0.02, 0.05\}$. Particle numbers: $N = 8$ (the $k = 0$ brane zero modes of both block types), $N = 688$ (the largest closed shell of the free $a_{4,0} = 0$ aufbau whose last level lies below the bulk edge 1.292292828068783; the bulk edge is an $L = 3$ value: 1.0977 at $L = 6$ and m as $L \to \infty$, and from $L = 3.5$ on the $N = 688$, $a_{4,0} = 0$ aufbau occupies $k = 0$ bulk levels, its KS gap 0.0452 closes and its energy does not converge up to $L = 6$) and $N = 136$ (the closed shell nearest 688/4). Couplings: λ_1 and λ_2 keep the first-order mean-field potential below 0.1 and 0.3 (units of m) along the whole history:

N	strength per unit lambda	lambda_1	lambda_2
8	5.138804	0.01946	0.05838
136	107.5483	0.0009298	0.002789
688	541.7126	0.0001846	0.0005538

This calibration is an $L = 3$ statement: the strength is the largest first-order mean-field potential per unit λ, which grows with L (for $N = 8$ it is the zero-mode potential at the tip, $\propto e^{4L}$), so the same rule applied at larger L drives λ_1 to zero: 1.2×10^{-7} ($N = 8$, falling like e^{-4L} from 0.01946 at $L = 3$), 1.2×10^{-7} ($N = 136$) and 1.0×10^{-11} ($N = 688$) at $L = 6$ (`Revision/kohn_sham/tip_convergence/`).

The canonical matrix holds 75 ground states (3 values of N, the five couplings $0, \pm\lambda_1, \pm\lambda_2$, five slices), each with its first excited state, its a_4 neighbours, its rescaling partner and, for $\lambda \neq 0$, its exact-Fock variant, and 135 Mermin states ($\lambda \in \{0, \pm\lambda_1\}$, three temperatures).

item	records
boundary conditions (Wolfram and sympy)	`bc_mirror_map_PA`, `bc_mirror_parities_of_densities`, `bc_brane_parity_conditions`, `bc_self_adjoint_boundary_term`, `bc_tip_family`, `bc_current_conserved_along_y`, `bc_exact_k0_spectra`, `bc_tip_asymptotics`, `brane_band_slope`
free problem (Rust solver)	`free_k0_analytic_spectra`, `free_zero_mode_exact`, `free_brane_band_slope`, `free_block_type_symmetries`, `free_particle_branch_labels`, `free_tip_angle_insensitivity`
free problem and parameters (reference)	`free_k0_analytic_spectra`, `free_convergence_order_two`, `free_k0_block_type_symmetry`, `free_zero_mode`, `free_brane_band_slope`, `free_particle_branch`, `parameters_closed_shells`, `parameters_calibration`, `richardson_asymptotic_ratio`, `richardson_uncertainty_validated`
parameters (cross-check)	`problem_definition_identical`, `parameters_particle_numbers`, `parameters_couplings`

6 Instantaneous ground and first excited states along the prescribed history

The history is $a_4 = AHx_4$ with $A = 1$ (the exponentially deflating member: the extra-time scale factor is $e^{-Hx_4} \sin^{1/6} z$), PRESCRIBED, not solved for (section 11). At every slice $a_{4,0}$ the ground state is the self-consistent aufbau state of the instantaneous problem; the first excited state is computed three ways: the Kohn-Sham gap $\varepsilon_{\mathrm{LUMO}} - \varepsilon_{\mathrm{HOMO}}$; the ensemble Δ-SCF energy (one

particle moved from the highest occupied degenerate group to the lowest empty group, spread uniformly over each group, self-consistent); and the list of the 24 lowest particle-hole excitations (`Revision/kohn_sham/results/excited/particle-hole/`). Every ground state of the matrix is closed-shell.

Ground and first excited states for $N = 136$ (`Revision/kohn_sham/results/ground/summary.csv` and `Revision/kohn_sham/results/excited/summary.csv`):

lambda	**a4,0**	**E_KS**	**KS gap**	**Delta-SCF**
0	0	80.28222	0.08267377	0.08267377
0	1	32.38412	0.03498894	0.03498894
0	2	12.44507	0.01445687	0.01445687
0.0009298	0	80.2837	0.08267139	0.08267139
0.0009298	1	32.39295	0.03497394	0.03497395
0.0009298	2	12.44706	0.01447131	0.01447131
-0.0009298	0	80.28074	0.08267616	0.08267615
-0.0009298	1	32.37559	0.03500291	0.0350029
-0.0009298	2	12.44277	0.01444421	0.01444422
0.002789	0	80.28666	0.08266659	0.0826666
0.002789	1	32.41162	0.03494039	0.03494042
0.002789	2	12.449	0.01450788	0.01450787
-0.002789	0	80.27779	0.0826809	0.0826809
-0.002789	1	32.35937	0.03502814	0.03502812
-0.002789	2	12.43783	0.01442269	0.01442269

The free Kohn-Sham gap along the history ($\lambda = 0$, `ground/summary.csv`):

N	**a4,0 = 0**	**a4,0 = 0.5**	**a4,0 = 1**	**a4,0 = 1.5**	**a4,0 = 2**
8	0.4307337	0.2726449	0.1703493	0.1050152	0.06415941
136	0.08267377	0.05383771	0.03498894	0.0226437	0.01445687
688	0.04518029	0.03145504	0.02048367	0.01330982	0.008610228

Reading (labelled interpretation of these numbers): as the extra times deflate, the 3-momenta redshift as $ke^{-a_{4,0}}$, the brane band flattens (its slope at $k = 0$ is c, proportional to $e^{-a_{4,0}}$, section 5), the Kohn-Sham gap falls for every N and E_{KS} falls for $N = 136$ and 688; for $N = 8$ (the $k = 0$ zero modes, on which $\kappa k = 0$) E_{KS} does not change with $a_{4,0}$. With $\lambda = 0$ the Δ-SCF energy equals the Kohn-Sham gap (worst difference 2.011e-11 over 15 cases, record `excited_delta_scf_free_equals_gap`); with interaction the two differ slightly, because the Δ-SCF state relaxes the mean field self-consistently.

item	Rust solver records	reference records	cross-check records
ground states	`ground_runs_completed`, `ground_scf_converged`, `ground_closed_shell`, `ground_window_complete`, `ground_N_conservation`, `ground_energy_two_forms`, `ground_orbital_boundary_conditions`	`ground_scf_converged`, `ground_grid_consistency`, `ground_closed_shell`, `ground_window_complete`, `ground_N_conservation`, `ground_energy_two_forms`	`ground_occupations_and_groups`, `ground_energies`, `ground_homo_lumo_gap`, `ground_eigenvalues`, `ground_profiles`
first excited state	`excited_delta_scf_free_equals_gap`, `excited_scf_converged`	`excited_delta_scf_free_equals_gap`, `excited_particle_hole_lowest_is_gap`	`excited_delta_scf`, `excited_particle_hole_lists`
exact-Fock variant	`exx_variant_scf_converged`, `exx_variant_y_conservation`	`exx_scf_converged`, `exx_N_conservation`, `exx_energy_two_forms`, `exx_emt_energy_integral`, `exx_y_conservation_integrated`	`exchange_delta_E_x`, `exx_variant_scf`

7 Adiabaticity and Fermi-level crossings

Exactly (mean field), $i\partial_{x_4}\chi = h(a_4(x_4))\chi$ conserves k, the block and the brane parity; transitions occur only between levels of the same $(k, j, \text{block}, \text{parity})$. The measure along $a_4 = AHx_4$ is

$$Q_{nm} = AH\,\frac{|\langle n|\partial_a h|m\rangle|}{(\varepsilon_n-\varepsilon_m)^2} = AH\,\frac{|\langle n|\partial_a m\rangle|}{|\varepsilon_n-\varepsilon_m|}, \qquad \partial_a h_j = -j\kappa k\sigma_3 + j(\partial_a M_{\text{eff}})\sigma_2 + \partial_a v_v,$$

for n occupied and m empty in the same sector; the state is adiabatic if $\max Q_{nm} \ll 1$, and Q_{nm}^2 is the leading-order transition probability. The self-consistent derivatives come from the states at $a_{4,0} \pm \delta$, $\pm 2\delta$ ($\delta = 2\times 10^{-3}$, Richardson); Hellmann-Feynman, $d\varepsilon_n/da = \langle n|\partial_a h|n\rangle$, is checked in 75 cases (worst 2.609e-9, record `adiabatic_hellmann_feynman`).

Results (`Revision/kohn_sham/results/adiabatic/adiabaticity.csv`): $Q_{\max} = 0$ exactly for $N = 8$ ($\partial_a h$ vanishes on the $k = 0$ sector), and the largest value over the whole matrix is 0.09345324 (state N688_lamm2_a00, pair `11:+1:even:0 -> 11:+1:even:1`: two levels of the same shell, block type and parity). For $\lambda = 0$:

N	a4,0	Q_max	delta eps of the pair	dE/da4 (EMT)	dE/da4 (differences)
136	0	0.08514877	2.05293	-71.43976	-71.43976
136	0.5	0.07455997	1.888369	-46.47084	-46.47084
136	1	0.06361841	1.745737	-30.11922	-30.11922
136	1.5	0.05274874	1.616341	-19.33966	-19.33966
136	2	0.03839473	1.524416	-12.18614	-12.18614
688	0	0.09345059	2.253405	-597.9157	-597.9157
688	0.5	0.08526	2.055018	-388.9333	-388.9333
688	1	0.0746905	1.890122	-253.016	-253.016
688	1.5	0.06373822	1.747327	-163.9948	-163.9948
688	2	0.05289328	1.617688	-105.4063	-105.4063

The last two columns are the energy change along the history, from the energy-momentum tensor ($dE/da_4 = -6\mathrm{Vol}_7 \int e^{6Hy}(p_3 - p_t)\, dy$, section 9) and from finite differences at fixed occupations.

Fermi-level crossings. When levels of different k cross the Fermi level between slices, the instantaneous aufbau ground state differs from the adiabatically continued state, and the solver flags it. No crossing occurs in the canonical matrix: every one of the 15 series of `Revision/kohn_sham/results/adiabatic/history.json` has an empty crossing list, and `adiabatic/fermi-level-crossings.csv` records no change of the occupied set. A demonstration at $\lambda = 0$ and $N = 696$ (the smallest closed shell of the $a_{4,0} = 0$ aufbau that occupies a level off the even $j = +1$ brane band; `adiabatic/crossing-demo.csv`):

a4,0	occupied set as at a4,0 = 0	open shell	E aufbau	E adiabatically continued	difference
0	true	false	690.7796	690.7796	0
0.5	false	true	444.1946	447.8513	3.65669
1	false	true	283.7106	289.7632	6.052645
1.5	false	true	179.4593	187.0712	7.611827
2	false	true	112.1016	120.725	8.623384

From $a_{4,0} = 0.5$ on, the $k = 0$ odd bulk level is emptied and the $n_2 = 12$ brane-band shell is filled: the instantaneous ground state is not the adiabatically reached state, which lies above it by the last column.

Interpretation (labelled): with $Q_{\max} \leq 0.09345324$ the instantaneous states are self-consistently adiabatic for the same-sector transitions at the rate $A = 1$ in the canonical matrix; this does not solve the time-dependent problem (OPEN), and where Fermi-level crossings occur (the demonstration) the instantaneous ground state is not the state that the evolution reaches.

Rust solver records	reference records	cross-check records
`adiabatic_hellmann_feynman`, `adiabatic_crossing_flag_demonstration`	`adiabatic_hellmann_feynman_discrete`, `crossing_demo_reference`	`adiabatic_dE_da4`, `adiabatic_Q_max`, `crossing_demonstration`

8 Mermin thermodynamics

The Mermin states at $T = 0.01, 0.02, 0.05$ solve the self-consistent problem at fixed N with μ the root of $\sum gf = N$ in a well-conditioned form (thermal particles above and holes below a split of the levels, `solver/src/mermin.rs`). The checks: Ω in its two forms, $-dF/dT = S_{\rm ent}$, $C_V = T\,dS_{\rm ent}/dT = dE/dT$ at fixed N, and the chemical potential against 40-digit roots on the solver's own final levels (40-digit and 50-digit roots agree to 7.5e-35, record `root_precision_self_check`). For $N = 136$, $\lambda = 0$ (`Revision/kohn_sham/results/thermo/thermodynamics.csv`):

a4,0	T	mu	E	entropy	Omega	sea holes / N
0	0.01	0.8339283	80.34462	7.763517	-33.14726	2.1e-56
0	0.02	0.8250861	80.74027	34.38332	-32.1591	9.49e-29
0	0.05	0.7941344	82.71202	91.45989	-29.86326	4.43e-12
2	0.01	0.1247136	12.99426	104.8261	-5.015053	1.29e-09
2	0.02	0.1150668	14.49922	205.4874	-5.259608	4.25e-05
2	0.05	0.05801333	23.38445	464.1395	-7.712339	0.112

Validity of the particle-only ensemble (diagnostic of the CONVENTION). For every thermal state the number of thermal holes that the excluded sea brane band would carry at the same μ and T is computed. It exceeds 1% of N in 15 of 135 states, all at $a_{4,0} \geq 1$; the largest is 30.976 N (record `thermo_sea_hole_diagnostic_computed`). There the particle-only Mermin ensemble is outside its range of validity and a thermal treatment of the sea (pairs) would be required; in the table above this is the row $a_{4,0} = 2$, $T = 0.05$ (state N136_lam0_a20_T50).

Rust solver records	reference records	cross-check records
`thermo_runs_completed`, `thermo_scf_converged`, `thermo_N_conservation`, `thermo_grand_potential_two_forms`, `thermo_entropy_identity`, `thermo_CV_identity`, `thermo_window_cut`, `thermo_window_shells_beyond`, `thermo_mu_well_conditioned_root`, `thermo_mu_vs_40digit_roots`, `thermo_sea_hole_diagnostic_computed`	`thermo_scf_converged`, `thermo_N_conservation`, `thermo_grand_potential_two_forms`, `thermo_CV_identity`, `thermo_entropy_identity`, `thermo_window_cut`, `thermo_sea_holes_tail`	`thermo_state_functions`, `thermo_derivatives`, `thermo_sea_hole_diagnostic`, `thermo_levels`, `thermo_mu_high_precision`, `thermo_mu_rounding_diagnostic`, `thermo_mu_rust_stated_bound`

Mermin-root report	records
`Revision/kohn_sham/reports/ks-rust-mermin-roots.json`	`solver_runs_completed`, `root_precision_self_check`, `solver_mu_within_rounding_bound`, `single_reproduces_committed_matrix`, `fixture_written`

9 Energy-momentum tensor profiles

With the sign convention of `Revision/SPEC.md` section 4 ($\rho = -T^{x_4}{}_{x_4}$, $p_\mu = T^{\mu}{}_{\mu}$) and $T^{\mu}{}_{\nu} = -\langle K^{\mu}{}_{\nu}\rangle + \delta^{\mu}_{\nu} e_{\rm int}$, the proper profiles are

$$\rho(y) = \sum wgf\,\varepsilon\, n_o - e_{\rm int}, \qquad p_3(y) = \sum wgf\,\tfrac{\kappa|k|}{3}\, t_o + e_{\rm int}, \qquad p_t(y) = e_{\rm int},$$
$$p_0(y) = \sum wgf\,\left[(\varepsilon - v_v)n_o - M_{\rm eff}\, s_o - \kappa|k|\, t_o\right] + e_{\rm int},$$

with $T^{x_4}{}_{y} = 0$ for eigen-orbitals. The extra-time pressure p_t has no kinetic part (good sector). Exact identities, checked for every state: the trace of the kinetic parts is $M_{\rm eff} S + v_v n$; conservation along the hidden direction, $p_0' + 6Hp_0 = 3H(p_3 + p_t)$; the energy change $\nabla_\mu T^{\mu}{}_{x_4} = -\partial_{x_4}\rho - 3a_4'(p_3 - p_t)$, i.e. $dE/da_4 = -6{\rm Vol}_7 \int_{-L}^{0} e^{6Hy}(p_3 - p_t)\,dy$; and $2{\rm Vol}_7 \int e^{6Hy}\rho\,dy = E_{KS}$. Integrated profiles ($2{\rm Vol}_7 \int_{-L}^{0} e^{6Hy} X\,dy$, `Revision/kohn_sham/results/ground/emt-integrals.csv`), $\lambda = 0$:

N	a4,0	int rho	int p3	int p__t	int p0
136	0	80.28222	23.81325	0	35.04349
136	0.5	51.24519	15.49028	0	26.62178
136	1	32.38412	10.03974	0	19.33023
136	1.5	20.20308	6.446555	0	13.173
136	2	12.44507	4.062048	0	8.472703
688	0	680.4412	199.3052	0	240.5598
688	0.5	437.5129	129.6444	0	188.2465
688	1	279.4249	84.33868	0	143.2073
688	1.5	176.7328	54.66494	0	104.3887
688	2	110.3867	35.13543	0	71.5579

The proper profiles on 151 points $y = -3 + 0.02i$ are in `Revision/kohn_sham/results/ground/profiles/` (one file per state). The energy density varies along the hidden direction in every state with a nonzero tensor (section 11). Equations of state seen by a 3-space observer are not part of this document; they are the work of `Revision/dark_sector/dirac16complex` (its Kohn-Sham history report `Revision/dark_sector/dirac16complex/reports/ks-history-run.json`: 5 of 5 PASS).

item	Wolfram and sympy records	Rust solver records	reference and cross-check records
tensor and identities	emt_orbital_components, emt_trace_identity, emt_y_conservation_orbital, emt_y_conservation_selfconsistent, emt_x4_component	emt_energy_integral, emt_trace_derivative_form, emt_y_conservation_integrated, emt_y_conservation_pointwise, emt_energy_change_dE_da4	emt_energy_integral, emt_y_conservation_integrated, emt_energy_change_dE_da4, emt_integrals, emt_brane_tip_values

10 The rescaling partners

Exact identity. Because $a_{4,0}$ enters only through $\kappa k = e^{-Hy}(ke^{-a_{4,0}})$,

$$\mathrm{KS}(a_{4,0};\, \Delta k, v_t, \lambda, N, T) = \mathrm{KS}(0;\, \Delta k\, e^{-a_{4,0}},\, v_t\, e^{-3a_{4,0}},\, \lambda, N, T)$$

for the levels, orbitals, proper densities and energy-momentum profiles. Reading (labelled): the slice $a_{4,0}$ of the deflating history is the slice 0 with a 3-torus larger by $e^{a_{4,0}}$ in each direction and an extra-time volume smaller by $e^{-3a_{4,0}}$; the proper 7-volume is the same, and the deflation of the extra times is exactly what keeps it so. The 3-momenta redshift while the proper box stays the same.

Computed. Each solver solves the partner independently (60 partners). The worst level difference is 1.478e-13 and the worst relative difference of E_{KS} and of the profiles 3.318e-13 (records `rescaling_identity_between_slices` and `rescaling_identity_energy_profiles`). For $N = 136$, $\lambda = \lambda_1$ (`Revision/kohn_sham/results/rescaling/rescaling.csv`):

a4,0	partner Delta k	partner v_t	max delta eps	relative delta E	partner E_KS
0.5	0.1516327	0.2231302	1.478e-13	7.099e-14	51.24916
1	0.09196986	0.04978707	1.101e-13	1.777e-14	32.39295
1.5	0.05578254	0.011109	8.749e-14	0	20.21303
2	0.03383382	0.002478752	7.661e-14	0	12.44706

Wolfram and sympy records	Rust solver records	reference and cross-check records
rescaling_identity	free_rescaling_relation_and_band_monotone, rescaling_identity_between_slices, rescaling_identity_energy_profiles	rescaling_identity_reference, rescaling_partners

11 The recorded states are not admissible sources of the a_4 equations

`Revision/SPEC.md` section 7 names the Kohn-Sham energy-momentum tensor as the source of the field equations for a_4. The Einstein-Lovelock equations of the author's metric (`Revision/field_equations_a4`) require of any source: (C1) every component independent of x_0; (C2) $p_3 + p_t = 2p_0$ (the algebraic identity $E^{x_1}{}_{x_1} + E^{x_5}{}_{x_5} = 2E^{x_0}{}_{x_0}$ of the Lovelock tensors); (C3) for the linear member $a_4 = AHx_4 + a_0$: $p_3 = p_t = p_0$ and constant ρ. The record `Revision/field_equations_a4/reports/ks-source-conditions.json` (6 of 6 PASS) finds:

1. Every one of the 70 nonzero ground-state profiles depends on x_0: $(\max\rho - \min\rho)/\max|T| \geq 0.0497329$ (record `ks_profiles_depend_on_x0`).
2. $\max_y |p_3 + p_t - 2p_0|/\max|T|$ lies between 2.09192 and 3.99006 (record `ks_profiles_violate_algebraic_condition`).
3. Integrated over the patch, $(\int p_3 + \int p_t)/(2\int p_0)$ differs from 1 for every nonzero state (closest 0.414328); for the history $N = 136$, $\lambda = 0$ it is 0.339767, 0.25969 and 0.239714 at $a_{4,0} = 0, 1, 2$ (record `ks_integrals_violate_algebraic_condition`): even an x_0-averaged source violates C2.
4. Along the history ρ changes with $a_{4,0}$ and $p_3 \neq p_t$, so C3 fails: the history is a PRESCRIBED background (record `ks_history_is_a_prescribed_background`).
5. Five states have an identically vanishing tensor (the free $N = 8$ zero modes), a zero source, for which Einstein gravity has no vacuum solution with $H > 0$ (record `ks_zero_source_states_listed`).
6. The verdict holds for every b_4 of the metric of 2026-10-14 (section 2.1): the diagonal left-hand sides stay free of x_0, so C1 is unchanged; the off-diagonal x_4-x_0 condition is not evaluated (OPEN) (record `ks_verdict_holds_for_every_b4`).

The full analysis `Revision/field_equations_a4/ks_source/reports/ks-source-a4.json` (23 of 23 PASS) refines this exactly: the states SATISFY the x_0 conservation identity, and $p_3 + p_t - 2p_0 = p_{0,y}/(3H)$ on shell, so C2 fails exactly because p_0 depends on x_0; at least a share 0.583068 of the energy density deviates from its hidden-direction average (L1 measure). Its conclusion: the coupled problem "metric of `Revision/SPEC.md` section 1 + Kohn-Sham source" has no solution, and nothing about a_4 is derived from it. In a stated approximation (the x_4 and $x_1 - x_5$ moments kept, the others dropped, with dropped residuals of order one) the Kohn-Sham source does not start or select the exponential deflation; it modifies a deflation set by the initial data, e.g. $a_4'(2)/a_4'(0) = 1.95362$, 1.1321 and 0.847549 for the source strengths $\sigma_0 = 10$, 1 and -1 of the series $N = 136$, $\lambda = 0$ in Einstein gravity, and with $\Lambda = 0$ and the initial rate $a_4'(0) = H$ the constraint forces $\kappa\bar{\rho}(0) < 0$, that is $\kappa < 0$ for this gas of positive energy density (a negative gravitational coupling in the sign convention of the energy-momentum tensor used by the field-equations record); then, for $N = 136$, $\lambda = 0$, the deflation halts ($a_4' = 0$) at $a_4 = 0.149623$ (Einstein), 0.0724197 (Einstein-Gauss-Bonnet) and 0.102772 (cubic Lovelock), where the integration stops (four of the ten series with 3-momentum were integrated; what follows the turning point is not computed). These are statements of that approximation, not of the field equations.

Consequence: the Kohn-Sham gas of this record is a test field on the prescribed background $a_4 = AHx_4$ without back-reaction; energies, pressures and equations of state along the history are not consequences of the coupled field equations.

report	records
`Revision/field_equations_a4/reports/ks-source-conditions.json`	`ks_profiles_depend_on_x0`, `ks_profiles_violate_algebraic_condition`, `ks_integrals_violate_algebraic_condition`, `ks_history_is_a_prescribed_background`, `ks_zero_source_states_listed`, `ks_verdict_holds_for_every_b4`
`Revision/field_equations_a4/ks_source/reports/ks-source-a4.json`	`A1_lovelock_identity_x1_plus_x5_equals_2x0`, `A6_x0_conservation_in_y`, `B0_wave1_report_verified`, `B2_x0_conservation_on_every_profile`, `B4_hidden_direction_mismatch`, `C1_averaged_algebraic_condition_fails`, `D3_lambda_zero_cases_halt`, `D5_einstein_sign_of_the_effect`
Kohn-Sham theory (the record and the label of the history)	`adiabatic_offdiagonal_identity`, `ks_theory_json_written`, `ks_theory_json_basis`, `ks_theory_json_slope`, `ks_theory_json_history_label`

12 The Kohn-Sham-level pairing T3

The block map of the chirality Γ, $(\chi, j) \to (\sigma_2\chi, -j)$, with the two brane parities exchanged and the tip angle $\theta \to \pi - \theta$, maps every self-consistent instantaneous Kohn-Sham state with (m, λ, θ) onto one with $(-m, +\lambda, \pi - \theta)$, with equal levels, occupations, Kohn-Sham energies and energy-momentum profiles and $S \to -S$ (ASSUMED Z2 brane, mean-field level). The theorem, its hypotheses and its proof are in `Revision/docs/PAIR_CREATION_PROOFS.md` section 8, verified by `Revision/pairing/kohn_sham/reports/wolfram-t3.json` (10 of 10 PASS) and `Revision/pairing/kohn_sham/reports/python-t3.json` (13 of 13 PASS); its completion record (`Revision/pairing/kohn_sham/t3-completion.json`) by `Revision/pairing/kohn_sham/reports/wolfram-t3-completion.json` (3 of 3 PASS) and `Revision/pairing/kohn_sham/reports/python-t3-completion.json` (7 of 7 PASS). The Rust solver contains a numerical self-test, labelled in its report as NOT a proof of T3: $(m, \lambda, \theta = 0)$ and $(-m, \lambda, \theta = \pi)$, solved independently, agree with opposite S (worst 9.95e-14, tolerance 1e-9; record `t3_block_map_solver_selftest`). T3 does not establish a creation process, a rate or an amplitude.

13 Verification records

13.1 Reports and their counts

Every check has a name, a verdict and a detail. Counts at the time of writing (the publication test re-reads them):

report	checks	PASS	FAIL
Revision/kohn_sham/reports/ks-theory-wolfram.json	46	46	0
Revision/kohn_sham/reports/ks-theory-python.json	58	58	0
Revision/kohn_sham/reports/ks-rust-solver.json	42	42	0
Revision/kohn_sham/reports/ks-rust-determinism.json	14	14	0
Revision/kohn_sham/reports/ks-rust-mermin-roots.json	5	5	0
Revision/kohn_sham/reports/ks-reference.json	37	37	0
Revision/kohn_sham/reports/ks-crosscheck.json	31	31	0
Revision/field_equations_a4/reports/ks-source-conditions.json	6	6	0
Revision/field_equations_a4/ks_source/reports/ks-source-a4.json	23	23	0
Revision/field_equations_a4/reports/wolfram-a4-report.json	64	64	0
Revision/field_equations_a4/reports/python-a4-report.json	63	63	0
Revision/pairing/kohn_sham/reports/wolfram-t3.json	10	10	0
Revision/pairing/kohn_sham/reports/python-t3.json	13	13	0
Revision/kohn_sham/tip_convergence/tip-convergence.json	7	7	0

13.2 The independent reference and the full cross-check

The reference shares no code with the Rust solver: different discretisation (staggered finite differences with Richardson extrapolation instead of RK4 shooting), its own SCF, its own parameter derivation (it re-derives the particle numbers and couplings from the stated rules) and its own uncertainty, validated on a fourth grid (record `richardson_uncertainty_validated`). The checker `Revision/kohn_sham/checker/crosscheck_ks.py` compares the two on the FULL canonical matrix: 75 ground states, 135 thermal states, 60 exact-Fock-variant states, 60 rescaling partners, 75 particle-hole lists and the 5 slices of the crossing demonstration, 162691 comparisons in all, under a tolerance rule fixed before the comparison,

$$|x_{\mathrm{Rust}} - x_{\mathrm{ref}}| \le 3\,(U_{\mathrm{ref}} + U_{\mathrm{Rust}}) + 10^{-12}\,\mathrm{scale}, \qquad U_{\mathrm{Rust}} = \tfrac{16}{15}\,|\mathrm{canonical} - \mathrm{refined}|,$$

with U_{ref} the reference's three-grid Richardson uncertainty and U_{Rust} measured for every state of the matrix by the refinement measurement, whose script and record are `Revision/kohn_sham/checker/measure_rust_refinement.py` and `Revision/kohn_sham/checker/rust-refinement.json`. All 31 checks of the cross-check pass:

check	check	check
inputs_all_pass	rust_refinement_applies_to_matrix	rust_refinement_applies_to_matrix_exx
problem_definition_identical	parameters_particle_numbers	parameters_couplings
ground_occupations_and_groups	ground_energies	ground_homo_lumo_gap
ground_eigenvalues	excited_delta_scf	excited_particle_hole_lists
emt_integrals	emt_brane_tip_values	ground_profiles
exchange_delta_E_x	adiabatic_dE_da4	adiabatic_Q_max
exx_variant_scf	rescaling_partners	crossing_demonstration
thermo_state_functions	thermo_derivatives	thermo_sea_hole_diagnostic
thermo_levels	thermo_mu_high_precision	thermo_mu_rounding_diagnostic
thermo_mu_rust_stated_bound	reference_outputs_lf_only	reference_manifest
reference_repeat_byte_identical		

13.3 Determinism and the refinement measurement

The Rust determinism report (14 of 14 PASS): a repeat of the canonical matrix is byte-identical in all 244 result files and in its check report (`repeat_byte_identical`, `repeat_report_byte_identical`, `outputs_lf_only`); a run with refined tolerances (900 to 1800 RK4 steps, root tolerance 1e-14, SCF tolerance 1e-12) agrees with the canonical run to 1.901e-12 in E_{KS} (relative), 2.135e-09 in 23724 levels, 2.800e-09 in the profiles, 9.024e-10 in $Q_{\max}$ and dE/da_4, 2.489e-10 in the thermodynamics and 5.689e-09 in C_V (`refined_same_inputs`, `refined_free_spectra_convergence_order`, `refined_ground_energies`, `refined_ground_homo_lumo_gap`, `refined_eigenvalues`, `refined_delta_sc`

f, refined_profiles, refined_adiabatic_derivatives, refined_thermodynamics, refined_mermin_root_path, refined_heat_capacity).

13.4 Re-verification for this document

For this document (2026-10-08) the committed Rust binary was run on the canonical matrix into a scratch directory (revision_ks_solver all --out <scratch>/rerun -report <scratch>/rerun-report.json): 42 of 42 checks PASS in 128 s of wall time on a machine shared with other jobs, and all 244 result files and the check report were byte-identical to the committed Revision/kohn_sham/results and Revision/kohn_sham/reports/ks-rust-solver.json. The other reports cited are the files of the working tree; their counts are re-read by the publication test.

14 Reproduction

From the repository root, in this order (the Kohn-Sham theory record reads the gammas of Revision/algebra; the Rust solver reads ks-theory.json; the Mermin-root tool runs after the canonical matrix; the cross-checker runs after the reference and the refinement measurement; the source-conditions checker reads the Kohn-Sham results and the Kohn-Sham theory checker reads its record; <scratch> is any empty directory outside the repository; the backslash is the line continuation of a POSIX shell):

```
wolframscript -file Revision/algebra/wolfram/verify_algebra.wls
python Revision/algebra/python/check_algebra.py
wolframscript -file Revision/field_equations_a4/wolfram/verify_field_equations_a4.wls
python Revision/field_equations_a4/python/check_field_equations_a4.py
wolframscript -file Revision/kohn_sham/theory/verify_ks_theory.wls
cargo build --release --manifest-path Revision/kohn_sham/solver/Cargo.toml
cargo test --release --manifest-path Revision/kohn_sham/solver/Cargo.toml
Revision/kohn_sham/solver/target/release/revision_ks_solver all \
    --out Revision/kohn_sham/results \
    --report Revision/kohn_sham/reports/ks-rust-solver.json
python Revision/kohn_sham/solver/tools/mermin_roots_mp.py --work <scratch>/mermin-roots
Revision/kohn_sham/solver/target/release/revision_ks_solver all \
    --out <scratch>/repeat --report <scratch>/repeat-report.json
Revision/kohn_sham/solver/target/release/revision_ks_solver all --refined \
    --out <scratch>/refined --report <scratch>/refined-report.json
python Revision/kohn_sham/solver/tools/compare_runs.py \
    --canonical Revision/kohn_sham/results \
    --canonical-report Revision/kohn_sham/reports/ks-rust-solver.json \
    --repeat <scratch>/repeat \
    --repeat-report <scratch>/repeat-report.json --refined <scratch>/refined \
    --refined-report <scratch>/refined-report.json \
    --report Revision/kohn_sham/reports/ks-rust-determinism.json
python Revision/kohn_sham/reference/run_reference.py --jobs 22 \
    --timing <scratch>/timing.json
python Revision/kohn_sham/checker/measure_rust_refinement.py \
    --work <scratch>/rust-single --jobs 22
python Revision/kohn_sham/checker/crosscheck_ks.py --work <scratch>/cc --jobs 22
python Revision/kohn_sham/tip_convergence/tip_convergence.py \
```

```
    --work <scratch>/tip --jobs 8
python Revision/field_equations_a4/python/check_ks_source_conditions.py
python Revision/kohn_sham/theory/check_ks_theory.py
python Revision/field_equations_a4/ks_source/ks_source_a4.py
python -m unittest Revision/field_equations_a4/ks_source/test_ks_source_a4.py -v
wolframscript -file Revision/pairing/kohn_sham/wolfram/verify_t3.wls
python Revision/pairing/kohn_sham/python/check_t3.py
python scripts/build_provenance_pdf.py Revision/docs/KOHN_SHAM_DEFLATING_FIELD.md \
    --developer-layout --specifications Revision/pdf-specifications.json
python -m unittest Revision/tests/test_kohn_sham_deflating_field_publication.py -v
```

Expected output: every verifier ends with all of its checks passing and writes the report of section 13.1 with the counts given there (the Wolfram Kohn-Sham verifier prints 46/46 checks passed; the Rust solver prints SUCCESS after 42 PASS lines; the cross-checker writes 31 of 31 PASS; the tip-cutoff study prints 7 PASS lines and SUCCESS); every run is deterministic, and a second run gives byte-identical outputs. Run times recorded in the READMEs of the folders (22 threads or processes; longer when the machine is shared): the Wolfram Kohn-Sham verifier about 40 to 85 s and the sympy checker about 75 to 165 s (`Revision/kohn_sham/theory/WOLFRAMSCRIPT_PROVENANCE.md`); the Rust canonical matrix 78.1 s on an idle machine (`Revision/kohn_sham/solver/README.md`; 128 s in the re-run of section 13.4), the repeat 178.4 s and the refined run 618.6 s on a shared machine; the reference 578.3 s (`Revision/kohn_sham/reference/README.md`); the refinement measurement 31.6 s idle and 85.2 s shared; the cross-checker 588.9 s idle and 721.2 s shared, most of it the repeat of the reference that it runs itself (`Revision/kohn_sham/checker/README.md`); the tip-cutoff study 541 s and 553 s in two complete runs with 8 workers on a shared machine (`Revision/kohn_sham/tip_convergence/README.md`); the Kohn-Sham source analysis 3 s to about 20 s (`Revision/field_equations_a4/ks_source/README.md`); the a_4 verifiers 43 to 70 s (Wolfram) and 7 to 28 s (sympy) (`Revision/field_equations_a4/README.md`); the T3 verifiers about 3 s and 1 s (`Revision/docs/PAIR_CREATION_PROOFS.md` section 9.3).

The PDF build runs the Markdown-to-LaTeX builder twice (with different hash seeds), runs pdflatex three times into each of two fresh directories, requires warning-free logs and byte-identical PDFs, and checks the PDF against its entry `kohn-sham-deflating-field` in `Revision/pdf-specifications.json` (the Revision registry; the registry of the earlier stages is not touched).

15 What is proved, computed, assumed and not established

15.1 Proved (exact, two independent verifiers)

1. The block reduction (section 3): for every history $a_4(x_4)$ the good-sector equation reduces exactly to eight 2×2 blocks h_j in the hidden coordinate; the spin-connection term $3H\gamma^0$ is removed by W^{-3}; the a_4' terms of the inflating and the deflating directions cancel.

2. The exchange functional (section 4): the exact local exchange of the uniform 8-fold good-sector gas at every temperature, $M_{\text{eff}} = m + \frac{15}{16}\lambda S$, $v_v = -\lambda n/16$; the exact Fock exchange of the closed-shell slab differs by $+\lambda Q^2/32$.

3. The boundary-value structure (section 5): self-adjointness with current-killing ends, the mirror map and the parities, the exact $k = 0$ spectra, the brane-band slope.

4. The rescaling identity (section 10), the energy-momentum tensor of the Kohn-Sham states with its trace and conservation identities (section 9), and the adiabaticity identity of the measure (section 7).

5. T3 (section 12; proved in `Revision/docs/PAIR_CREATION_PROOFS.md`), under its stated hypotheses.

15.2 Computed (numerical, two independent solvers, cross-checked)

1. The instantaneous ground states, first excited states (gap, Δ-SCF, particle-hole lists), Mermin states, energy-momentum profiles, rescaling partners and exact-Fock variants of the canonical matrix (sections 6 to 10), with the Rust solver and the reference agreeing on the full matrix within the stated uncertainty (31 of 31 checks, 162691 comparisons).

2. The adiabaticity measure ($Q_{\max} \leq 0.09345324$) and the absence of Fermi-level crossings in the canonical matrix; the crossing demonstration at $N = 696$ (section 7).

3. That the recorded states are not admissible sources of the a_4 equations (section 11).

4. The dependence on the tip cutoff L from 3 to 6 of 29 recorded states, the 27 ground states with $N \in \{8, 136, 688\}$, $\lambda \in \{0, \pm\lambda_1\}$, $a_{4,0} \in \{0, 1, 2\}$ and two Mermin states of $N = 136$ ($\lambda = 0$, $a_{4,0} = 2$, $T = 0.05$; $\lambda = \lambda_1$, $a_{4,0} = 1$, $T = 0.02$) (section 5; `Revision/kohn_sham/tip_convergence/`, 7 of 7 checks PASS; the Rust solver at every L, the reference at $L = 3$ and 4).

15.3 Assumed or chosen

1. The Z2 brane (orbifold) at the patch end $z = \pi/2$ (ASSUMED); the metric is degenerate there and no junction condition is derived.

2. The regular tip at the cutoff $L = 3$ with $\theta = 0$ (chosen). Its effect is measured (`Revision/kohn_sham/tip_convergence/`, $L = 3$ to 6; section 5): the recorded free results, except $N = 688$ at $a_{4,0} = 0$, lie within 1.4% (E_{KS}) and 12% (the p_0 integral) of their large-L values; the interaction effects do not: the proper zero-mode density grows toward the tip like $e^{(6H-2m)|y|}$, the recorded $N = 8$ interaction energy is about one third of its large-L value, and the interaction shift $E_{KS}(\pm\lambda_1) - E_{KS}(0)$ of $N = 136$ at $a_{4,0} = 2$ is at large L about 28 ($+\lambda_1$, value at $L = 5.5$) and 21 ($-\lambda_1$, extrapolated) times its $L = 3$ value. The couplings (calibrated at $L = 3$) and $N = 688$ (bulk edge at $L = 3$) are themselves $L = 3$ choices.

3. The filling convention (particles on the positive branch and on the $k = 0$ zero modes; the sea not populated thermally), whose justification is OPEN; the sea-hole diagnostic shows where it fails (15 of 135 thermal states).

4. The history $a_4 = AHx_4$, $A = 1$, PRESCRIBED, not solved for.

5. The parameters of section 5 (N, λ, L, Δk, v_t, T).

15.4 Not established

1. The time-dependent (non-adiabatic) Kohn-Sham problem (TDDFT) is OPEN: only instantaneous states are computed; the measure $Q_{\max}$ is a consistency indicator, not a solution of the evolution.

2. Correlation: the functional is Hartree plus exchange only; no correlation functional is derived or used.

3. Back-reaction: the Kohn-Sham states are a test field on a prescribed background; no recorded state is an admissible source of the field equations for a_4, and no history $a_4(x_4)$ is derived from the Kohn-Sham source.

4. Equations of state and the dark-sector hypothesis are not part of this document (`Revision/dark_sector/dirac16complex`).

5. No creation process, rate or amplitude for universes follows from the Kohn-Sham model or from T3.

6. A thermal treatment of the Dirac sea (pairs) where the particle-only ensemble fails; states beyond the computed range $a_{4,0} \in [0, 2]$; the continuum limit $\Delta k \to 0$. The limit $L \to \infty$ is established (numerically, with uncertainties) only for the states and quantities marked converged or with a geometric tail in `Revision/kohn_sham/tip_convergence/README.md`; it is NOT established for $N = 688$ at $a_{4,0} = 0$ (the $k = 0$ bulk levels approach m algebraically and the occupied set keeps changing), nor for the 7 interacting states whose successive differences of E_{KS} do not meet the geometric-tail rule before the self-consistent iteration fails (it fails at some larger L for 13 of the 19 studied; for $N = 8$ see section 5). It is not computed at $L \neq 3$ for λ_2, the slices 0.5 and 1.5, $T = 0.01$ and the other thermal states, the excited states and the adiabaticity measure, nor at $L > 6$. The record's calibration rule drives λ_1 to zero as L grows.